HRISHIKESH BELAGALI

Quantum Algorithms & Simulation Researcher

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RESEARCH INTERESTS

- Classical simulation of quantum circuits
- Quantum algorithms for chemistry
- Quantum advantage
- Fault-tolerant quantum compiling
- Quantum error correction

SKILLS

Languages: Python, Rust, C++, Tcl, LaTeX

Packages: Qiskit, ffsim, Cirq, QuTiP, JAX, NumPy, SciPy, PyTorch, pyo3

Methods: Pauli/Majorana propagation, fermionic circuit simulation, Clifford+T synthesis, tensor networks, GPU/HPC computing

EDUCATION

Computer Science (B.S) — **Michigan State University Honors College**
2024-2027 (expected)
Current GPA: 4.0

RELEVANT COURSEWORK

Quantum Algorithms†, Tensor Networks†, Quantum Physics, Computational Foundations of Artificial Intelligence†, Linear Algebra

†**Graduate course**

PUBLICATIONS

[1] **Hrishikesh Belagali**, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, "Efficient classical simulation of large-scale unitary cluster Jastrow circuits," *arXiv:2607.21337 (2026)*.
Submitted to *Physical Review Letters* (revised manuscript under review).

[2] **Hrishikesh Belagali**, Ryan LaRose. "Propaq: A Python package for Heisenberg propagation", *arXiv:2609.07730 (2026)*.
Submitted to *ACM Transactions on Software Engineering and Methodology*.

[3] **Hrishikesh Belagali**, Nitin Kumar Singh, Josh V. Vermaas, "The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*," chemRxiv:15004463 (2026).
Submitted to *ACS Journal of Physical Chemistry B* (under revision).

EXPERIENCE

2026– **Undergraduate Research Assistant** — **MSU-Q**

- Developed fermionic backpropagation, an efficient classical method for simulating large-scale unitary cluster Jastrow circuits, combining new analytical reductions with GPU-accelerated implementation to achieve a 70-fold speedup for circuits with up to 77 qubits, and 3,500 two-qubit gates.
- Implemented fermionic backpropagation in Qiskit Community's *ffsim*, a Python package for simulating fermionic quantum systems, extending the method's accessibility to the broader quantum chemistry research community.
- Built *propaq*, a Heisenberg propagation package achieving runtime competitive with state-of-the-art simulators while offering greater customizability, enabling circuit simulations in custom bases with support for custom noise and truncation policies.

2025–2026 **Honors Research Scholar** — **MSU Honors College/MSU-Q**

- Developed a gateset-rotation algorithm for identifying unitary basis transformations to reduce non-Clifford synthesis cost in fault-tolerant quantum circuits.
- Investigated the utility of unitary basis transformations for Hamiltonian pre-processing in time evolution simulations.

2024– **Undergraduate Research Assistant** — **Vermaas Lab, Plant Research Laboratory**

- Ran large-scale molecular dynamics simulations of bacterial membranes on HPC clusters (40,000+ GPU hours, 30TB+ of data) using NAMD.
- Built Python/Tcl analysis pipelines identifying the membrane components responsible for alcohol tolerance.

CONFERENCES

2026 *MQC, IN*
Efficient classical simulation of large-scale unitary cluster Jastrow circuits
Poster presented at MQC Entanglement 2026

2026 *MSU UURAF*
Reducing T Gate Counts via Gateset Rotation
Poster presented at MSU Undergraduate Research and Arts Forum. **Awarded first place** in the Engineering, Computer Science and Mathematics division.

2026 *ACS, IL*
Molecular mechanisms of alcohol tolerance in Clostridium acetobutylicum and Zymomonas mobilis: The role of cyclopropane fatty acids and hopanoids **acs.digitellinc.com**
Poster presented at ACS Fall 2026 Conference

2026 *GLBRC, WI*
The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*
Poster presented at Great Lakes Bioenergy Research Center Annual Science Meeting.

RESEARCH SOFTWARE & OPEN SOURCE

propaq
Python, Rust
propaq **github.com/pypi.org**
Creator and primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.
- Implemented thread-safe kernels and compact representation of Pauli and Majorana strings in Rust for efficient multi-threaded simulation.
- Developed a Python interface integrating *propaq* with Qiskit and Cirq for seamless simulation workflows using pyo3 bindings.
- Designed a plugin system for developing custom noise and truncation policies in Python, Rust, C, and Julia to enable user customization with minimal overhead.

ffsim
Python, Rust
ffsim **github.com**
Contributed to *ffsim*, a Python package for simulating fermionic quantum systems.
- Implemented fermionic backpropagation, from "Efficient classical simulation of large-scale unitary cluster Jastrow circuits" (arXiv:2607.21337) into *ffsim* for fast unitary cluster Jastrow ansatz simulation.
- Created a JAX-based implementation for GPU-accelerated variational optimization.

QuTiP
Python
Quantum Toolbox in Python (QuTiP) **github.com**
Contributed to QuTiP, an open-source software for simulating quantum systems.
- Implemented functionality to return eigenkets as an operator in the Qobj class.

tce-lib
Python
tce-lib **github.com**
Contributed to tce-lib, a Python package for building tensor cluster expansion models for concentrated alloys.
- Implemented lazy evaluation of Monte Carlo trajectories for efficient simulation on HPC clusters.

RESEARCH COMMUNICATION

2026 *MSU CoE, MI*
"Student View: Challenging the Quantum Advantage Boundary" **engineering.msu.edu**
Article highlighting my quantum computing research journey, published by the College of Engineering, Michigan State University.

HONORS AND AWARDS

- First place at MSU UURAF in the Engineering, Computer Science and Mathematics division (2026)
- Wielenga Research Scholarship, Honors College (2025-2026) – Awarded to 15-20 MSU students per year
- Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024) – $2,000
- MSU President's Scholarship (2024) – $30,000

SELECTED PROJECTS

Python
Qiskit
Adiabatic Quantum Computation **github.com**
Implemented an adiabatic algorithm in Qiskit for QUBO problems via Trotterized Ising-model evolution; extended it in Graduate Quantum Algorithms to study hardware-noise effects on accuracy and convergence.